Stat 5200, Module 4: Factorial Design and Analysis

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Overview

Objectives and overview

- This material is based on Chapters 8–10 of the textbook "A First Course in Design and Analysis of Experiments" (Oehlert, 2000).
- Module outcomes include:
  - You will understand complete and incomplete factorial treatment structures.
  - You will develop an intuitive and formal understanding of interaction effects.
  - You will be able to specify and fit additive and non-additive two-way ANOVA models.
  - You will learn how to construct and interpret main effect contrasts and interaction contrasts.
  - You will know how to fit and analyze factorial designs in R using `aov()` and `emmeans()`.

Factorial Treatment Structure

Introduction to Factorial Structures

- In many experiments, we want to evaluate the effect of two or more explanatory factors simultaneously.
- A factorial treatment structure defines treatments as combinations of levels of each factor.
- Suppose Factor A has a levels and Factor B has b levels.
- A complete factorial design includes all $g = a \times b$ possible treatment combinations.

Introduction to Factorial Structures II

Example 1: 2×3 Diet Experiment

- Factor 1: Protein Source (Beef, Cereal, Pork) $\implies b = 3$.
- Factor 2: Protein Intake (High, Low) $\implies a = 2$.
- Total treatment combinations: $g = 2 \times 3 = 6$ unique diets.

Introduction to Factorial Structures III

Example 2: 2×2 Fertilizer Experiment

- Factor A: Superphosphate (P) $\rightarrow$ Levels: {Not Added, Added}
- Factor B: Potash (K) $\rightarrow$ Levels: {Not Added, Added}
- $g = 2 \times 2 = 4$ treatments:
  1. None (P_1K_1)
  2. Superphosphate only (P_2K_1)
  3. Potash only (P_1K_2)
  4. Both Superphosphate and Potash (P_2K_2)

Main Effects and Interactions

- We are generally interested in both the *main effects* of each treatment, but also the *interactions* between treatments. Consider the example below (Example 2):

	Beef	**Cereal**	**Pork**	**Marginal Row Mean**
High Intake	$\mu_{B,H}$	$\mu_{C,H}$	$\mu_{P,H}$	μ_H
Low Intake	$\mu_{B,L}$	$\mu_{C,L}$	$\mu_{P,L}$	μ_L
Marginal Col Mean	μ_B	$\bar{\mu}_C$	μ_L	μ (Grand Mean)

Main Effects and Interactions II

- **Main Effects:** Evaluated by comparing the marginal row means ($\bar{\mu}_{1\bullet}$ vs. $\bar{\mu}_{2\bullet}$) or the marginal column means ($\bar{\mu}_{\bullet 1}$, $\bar{\mu}_{\bullet 2}$, $\bar{\mu}_{\bullet 3}$). For example, what is the difference in response between high and low intake treatments?
- **Interactions:** Evaluated by looking at differences between cells. For example, is the difference between High and Low intake the same for Beef as it is for Cereal?

$$(\mu_{B,H} - \mu_{B,L}) \stackrel{?}{=} (\mu_{C,H} - \mu_{C,L})$$

Main Effects and Interactions III

- While we *could* formalize and test all these specific comparisons using individual contrasts, this can be tedious and less interpretable in some cases.
- Instead, we will introduce the Two-Way ANOVA as an efficient way to test for overall main effects and interactions simultaneously, with highly interpretable results.
- Like one-way ANOVA, we will express the total sum of squares in terms of treatments and errors.

Notation for Two-Way Factorials

- To identify observations in a two-way design, we use three subscripts:

$$y_{ijk}$$

  - $i = 1, \ldots, a$: level of Factor A
  - $j = 1, \ldots, b$: level of Factor B
  - $k = 1, \ldots, n$: replication number within treatment combination (i, j)
- Summary Notation:
  - Sample mean for treatment (i, j): $\bar{y}_{ij\bullet} = \frac{1}{n} \sum_{k=1}^{n} y_{ijk}$
  - Marginal mean for level i of Factor A: $\bar{y}_{i\bullet\bullet} = \frac{1}{bn} \sum_{j=1}^{b} \sum_{k=1}^{n} y_{ijk}$

Notation for Two-Way Factorials II

- Marginal mean for level j of Factor B: $\bar{y}_{\bullet j \bullet} = \frac{1}{an} \sum_{i=1}^{a} \sum_{k=1}^{n} y_{ijk}$
- Overall grand mean: $\bar{y}_{\bullet\bullet\bullet} = \frac{1}{abn} \sum_{i=1}^{a} \sum_{j=1}^{b} \sum_{k=1}^{n} y_{ijk}$

The Two-Way ANOVA Model

Model Specification

- For completely randomized designs, we will start with balanced factorial treatments. All $n_{ij} = n$ for all i, j.
- We model the response y_{ijk} as:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where:

- μ: Overall grand mean.
- α_i: Main effect of level i of Factor A.
- β_j: Main effect of level j of Factor B.
- $(\alpha\beta)_{ij}$: Interaction effect between level i of A and level j of B.
- $\epsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2)$: Random errors.

Model Specification II

- Parameter Identification Restrictions (Sum-to-Zero Constraints):

$$\sum_{i=1}^{a} \alpha_i = \sum_{j=1}^{b} \beta_j = \sum_{i=1}^{a} (\alpha\beta)_{ik} = \sum_{j=1}^{b} (\alpha\beta)_{kj} = 0 \text{ for all } k$$

- Hierarchy of Models:
  1. **Full (Non-additive) Model**:
     $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$
  2. **Additive Model**: $y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk}$
  3. **Main Effect Models**: $y_{ijk} = \mu + \alpha_i + \epsilon_{ijk}$,
     $y_{ijk} = \mu + \beta_j + \epsilon_{ijk}$
  4. **Constant Model**: $y_{ijk} = \mu + \epsilon_{ijk}$

Degrees of Freedom

- The degrees of freedom for each main effect is computed as we would in a one-way ANOVA:
  - Main effect A: $(a-1)$
  - Main effect B: $(b-1)$
  - Interaction effect (AB): $(a-1)(b-1)$.

Understanding Interaction Effects

What is an Interaction Effect?

- Definition: An interaction effect occurs when the effect of one factor on the response depends on the level of another factor.
- Ask yourself: *"Does the difference between levels of Factor A stay the same regardless of Factor B?"*
  - If **YES**: The factors are additive (No Interaction).
  - If **NO**: There is an interaction between factors.

What is an Interaction Effect? II

Intuition: "Difference of Differences"

Consider a 2×2 design with population means μ_{ij}:

- Simple main effect of A when Factor B is at level 1: $\mu_{11} - \mu_{21}$
- Simple main effect of A when Factor B is at level 2: $\mu_{12} - \mu_{22}$

The interaction effect is the difference between these two simple effects:

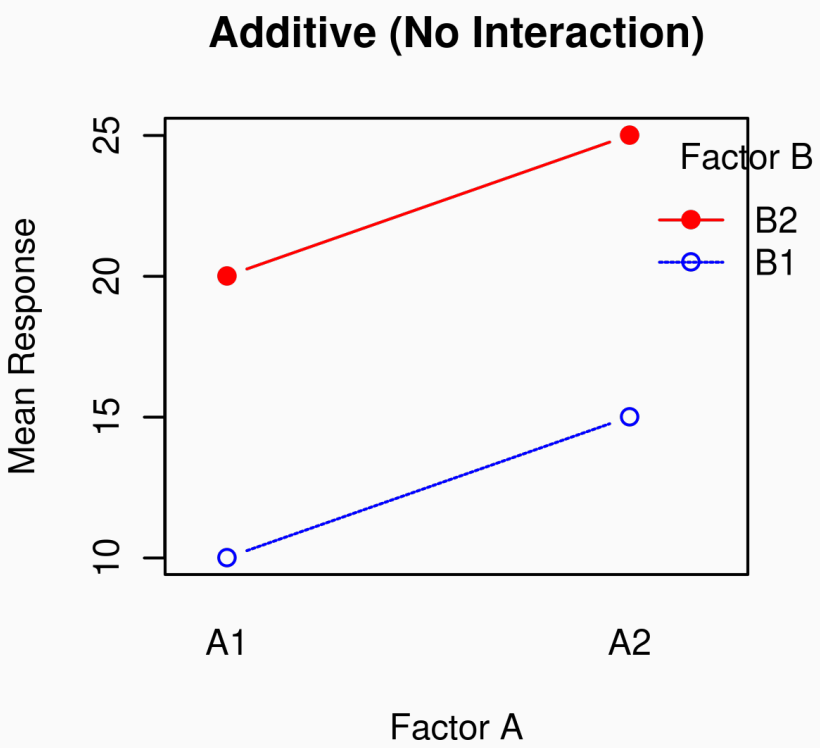
$$(\mu_{11} - \mu_{21}) - (\mu_{12} - \mu_{22})$$

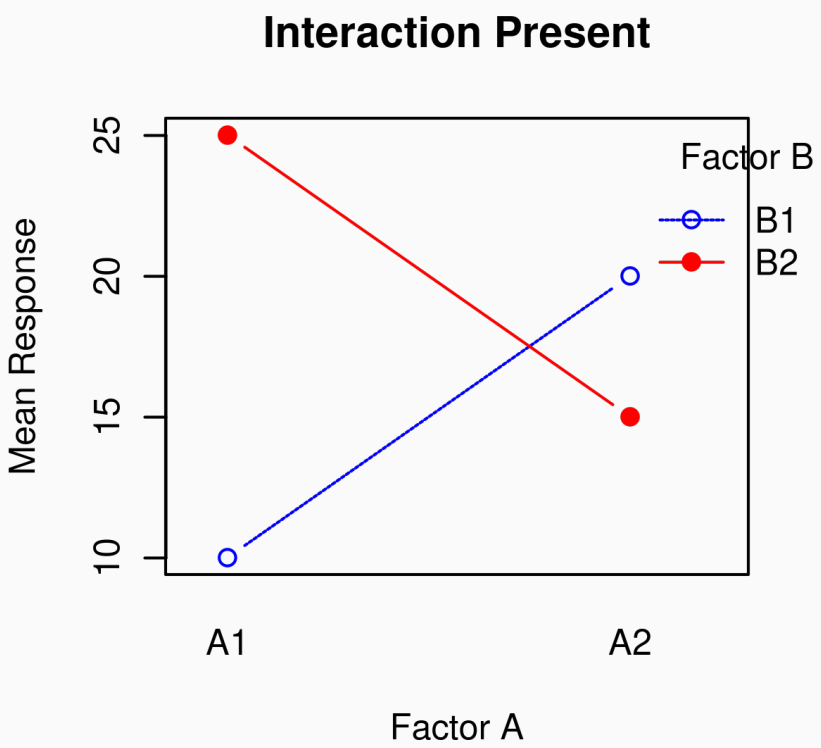
If this difference equals zero, the effect of A is constant across levels of B. If non-zero, an interaction exists.

Graphical Diagnosis: Interaction Plots

- An interaction plot displays treatment group means across levels of one factor, with lines connecting levels of the second factor.
- **Parallel lines** $\Longrightarrow$ No Interaction (Additive).
- **Non-parallel or crossing lines** $\Longrightarrow$ Interaction present.

Graphical Diagnosis: Interaction Plots II





ANOVA Mechanics and Hypothesis Testing

Two-Way ANOVA Table (Balanced Design)

- For a balanced design (n observations per cell, $N = abn$ total):

Source	DF	SS	MS	F-Statistic
Factor A	$a-1$	SS_A	MS_A	$F_A = \frac{MS_A}{MS_E}$
Factor B	$b-1$	SS_B	MS_B	$F_B = \frac{MS_B}{MS_E}$
Interaction	$(a-1)(b-1)$	SS_{AB}	MS_{AB}	$F_{AB} = \frac{MS_{AB}}{MS_E}$
Error	$ab(n-1)$	SS_E	MS_E	
Total	$abn-1$	SS_T		

See compiled notes for detailed formulas.

Two-Way ANOVA Table (Balanced Design) II

Sequence of Testing: Top-Down Rule

1. **Step 1: Test Interaction ($H_0 : (\alpha\beta)_{ij} = 0$ for all i, j)**
   - If **Significant**: Stop! The model is non-additive. Do not interpret main effect F-tests in isolation. Examine simple main effects or interaction contrasts.
2. **Step 2: Test Main Effects**
   - If interaction is non-significant, proceed to test $H_0 : \alpha_i = 0$ and $H_0 : \beta_j = 0$ independently using the main effect F-tests.

Applied Examples in R

Example 1: Significant Interaction (ToothGrowth)

- The built-in `ToothGrowth` dataset measures tooth length (`len`) in guinea pigs under two factors:
  - **Supplement (`supp`):** Orange Juice (OJ) or Ascorbic Acid (VC).
  - **Dose (`dose`):** 0.5, 1.0, or 2.0 mg/day.
- First, we must convert `dose` from a numeric variable into a factor so R treats it as categorical.

```r
# Load data and convert dose to a factor
data(ToothGrowth)
ToothGrowth$dose <- as.factor(ToothGrowth$dose)
```

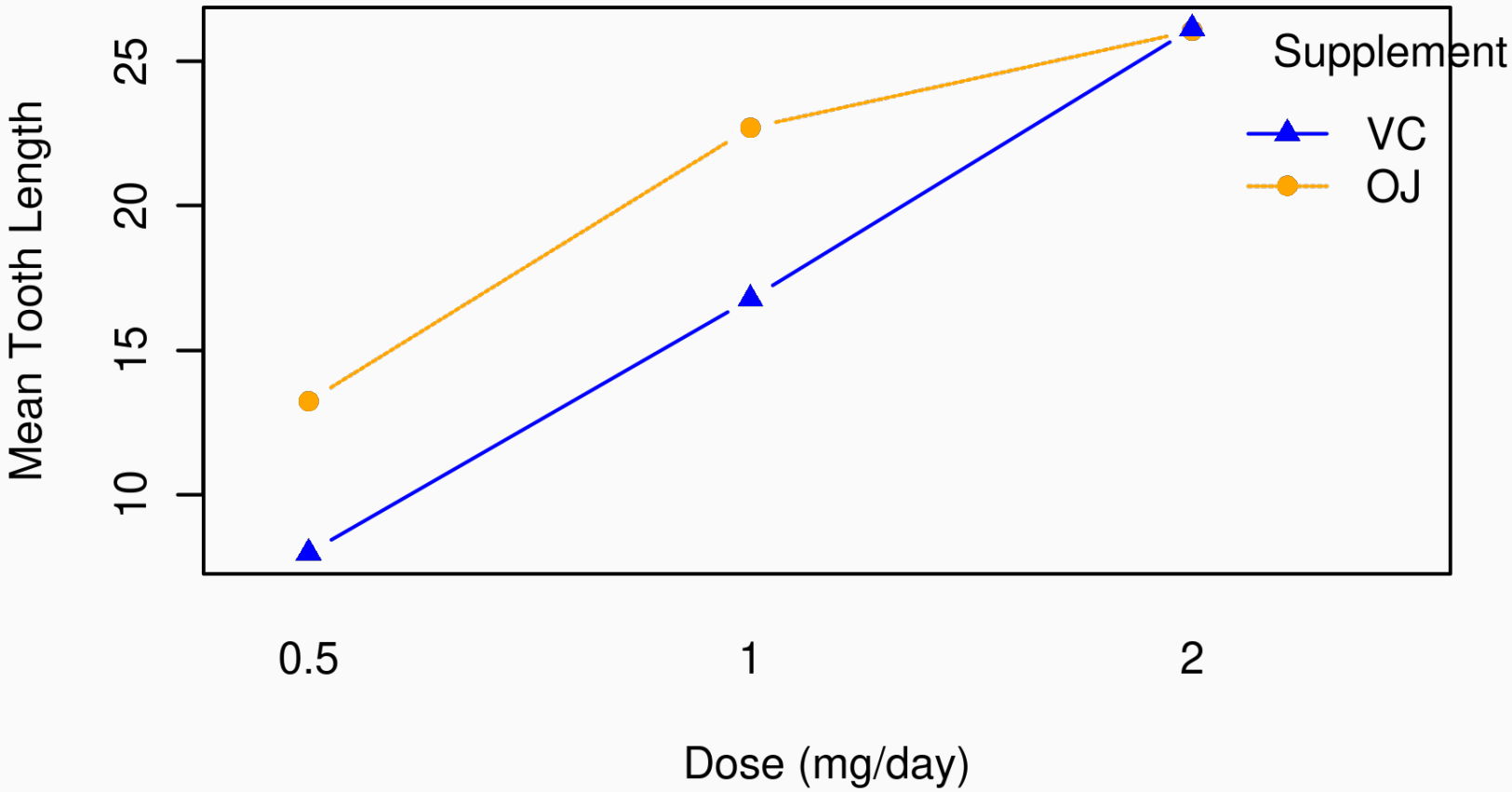
Example 1: Significant Interaction (ToothGrowth) II

Visualizing the Interaction

We start with an interaction plot. Crossing or converging lines suggest an interaction effect.

```r
interaction.plot(
  x.factor = ToothGrowth$dose,
  trace.factor = ToothGrowth$supp,
  response = ToothGrowth$len,
  type = "b", pch = c(16, 17),
  col = c("orange", "blue"),
  xlab = "Dose (mg/day)",
  ylab = "Mean Tooth Length",
  trace.label = "Supplement"
)
```

Example 1: Significant Interaction (ToothGrowth) III



Example 1: Significant Interaction (ToothGrowth) IV

Fitting the Model and Testing (Top-Down Rule)

We fit the full model with the * operator and check the ANOVA table.

```
mod_tooth <- aov(len ~ supp * dose, data = ToothGrowth)
summary(mod_tooth)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
supp          1  205.4   205.4  15.572 0.000231 ***
dose          2 2426.4  1213.2  92.000  < 2e-16 ***
supp:dose     2  108.3    54.2   4.107 0.021860 *
Residuals    54  712.1    13.2
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 1: Significant Interaction (ToothGrowth) V

- **Interpretation:** The `supp:dose` interaction is significant ($p = 0.02$). Following our Top-Down Rule, we **stop** and do not interpret the main effects blindly. The effect of the supplement depends entirely on the dose.

Example 1: Significant Interaction (ToothGrowth) VI

Analyzing Simple Effects with `emmeans`

Because the interaction is significant, we should look at the simple effects – pairwise differences that vary just a single factor. In this case, the effect of the supplement *at each specific level* of the dose.

Example 1: Significant Interaction (ToothGrowth) VII

```
library(emmeans)
# The "|" operator tells emmeans to compare 'supp' AT each level
emm_tooth <- emmeans(mod_tooth, ~ supp | dose)
pairs(emm_tooth)

dose = 0.5:
 contrast estimate   SE df t.ratio p.value
 OJ - VC      5.25 1.62 54   3.233  0.0021

dose = 1:
 contrast estimate   SE df t.ratio p.value
 OJ - VC      5.93 1.62 54   3.651  0.0006

dose = 2:
 contrast estimate   SE df t.ratio p.value
 OJ - VC     -0.08 1.62 54  -0.049  0.9609
```

Demystifying the `emmeans` Syntax

- In a one-way ANOVA, we could use `emmeans(model, ~ Factor)` to get the estimated means for each group.
- In a two-way ANOVA with a significant interaction, we must examine simple main effects. We need to know the effect of one factor *at specific levels* of the other factor.

Demystifying the `emmeans` Syntax II

Breaking down: `~ supp | dose`

- `~ supp`: This tells R to calculate the differences between the levels of the Supplement factor (OJ vs. VC).
- `|` **(The Pipe/Bar)**: In R formulas, the vertical bar means "given" or "conditioned on."
- `dose`: This is the grouping variable. It tells R to split the analysis up by the levels of Dose.

Demystifying the `emmeans` Syntax III

- **Plain English Translation:** "Compare the mean tooth length of OJ vs. VC, and do this comparison three separate times: once for the 0.5 dose, once for the 1.0 dose, and once for the 2.0 dose."

Demystifying the `emmeans` Syntax IV

- From our figure of the data (interaction plot), the Tooth Growth data is an example we may want to consider polynomial contrasts for factorial data. **There will be a 6000-level HW question** assignment related to this.

Example 2: No Interaction (Additive Model)

- The `npk` dataset in Canvas records pea yields based on applying Nitrogen (`N`) and Phosphate (`P`).
- Let's evaluate if Nitrogen and Phosphate interact to affect `yield`.

```
NP <- read.csv('NP.csv')
str(NP)

'data.frame': 24 obs. of  3 variables:
 $ yield: num  49.5 62.8 46.8 57 59.8 58.5 55.5 56 62..
 $ P    : int  1 1 0 0 0 1 0 1 1 1 ...
 $ N    : int  0 1 0 1 1 1 0 0 0 1 ...
```

Example 2: No Interaction (Additive Model) II

```
mod_NP <- aov(yield ~ N * P, data = NP)
summary(mod_NP)

            Df Sum Sq Mean Sq F value Pr(>F)
N            1  189.3  189.28   5.758 0.0263 *
P            1    8.4    8.40   0.256 0.6187
N:P          1   21.3   21.28   0.647 0.4305
Residuals   20  657.4   32.87
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

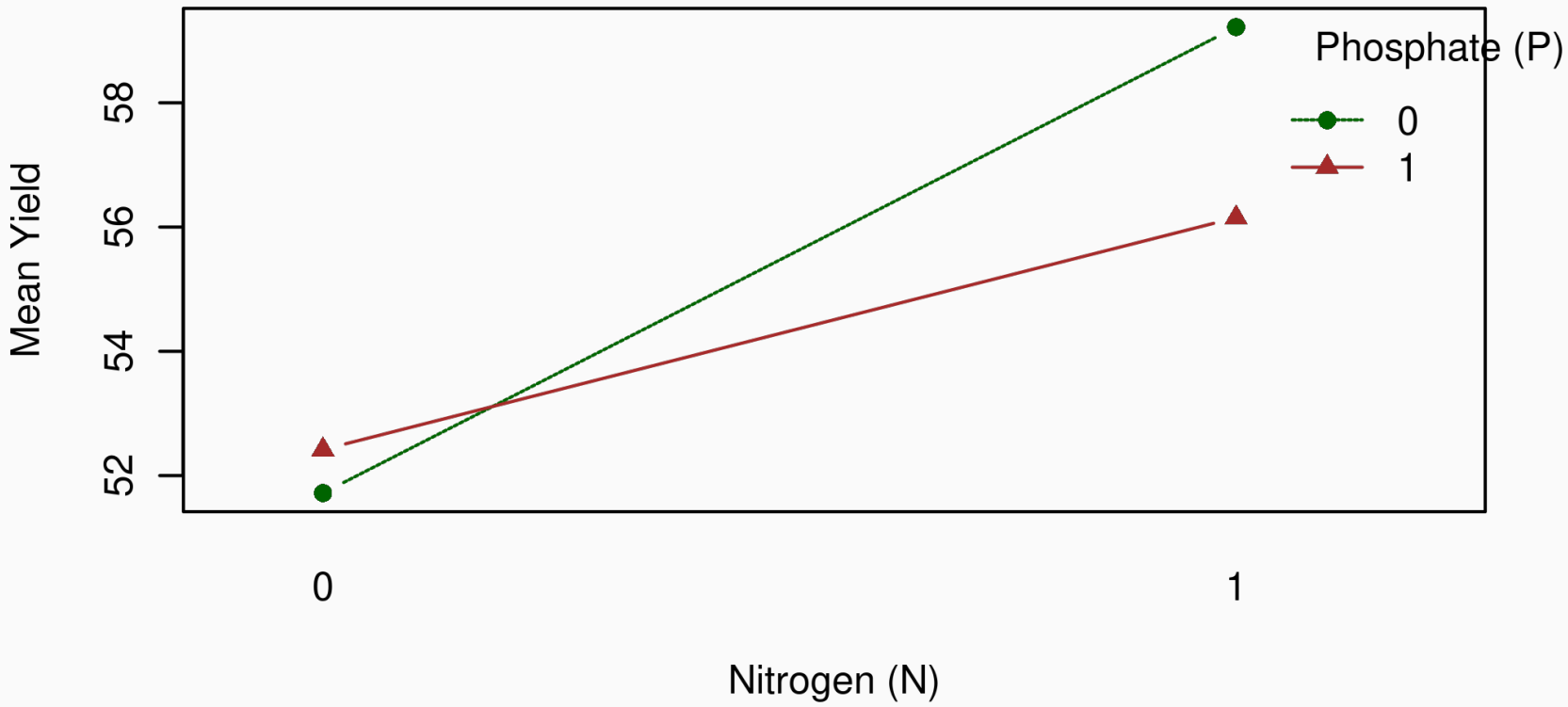
Example 2: No Interaction (Additive Model) III

- **Interpretation:** The `N:P` interaction is not significant ($P = 0.4305$). That is, there is not strong evidence in the data that the effect of Nitrogen inclusions on Mean Yield depends on the inclusion of Phosphate.
- We can examine this further with the interaction plot

Example 2: No Interaction (Additive Model) IV

```r
# Visualize the data
interaction.plot(
  x.factor = NP$N,
  trace.factor = NP$P,
  response = NP$yield,
  type = "b", pch = c(16, 17),
  col = c("darkgreen", "brown"),
  xlab = "Nitrogen (N)", ylab = "Mean Yield",
  trace.label = "Phosphate (P)"
)
```

Example 2: No Interaction (Additive Model) V



Example 2: No Interaction (Additive Model) VI

- Not significant interaction. Now what?
  - Historically, we might want to re-fit an additive model to conserve degrees of freedom.
    Problem: This is called *post-selection inference*, and can mean we have inflated type-I error rates.
  - Current recommended practice: we are more concerned about the increased Type-I error rate than we are of improving statistical power. We will interpret P-values in the context of accounting for interactions, even if they are non-significant.

Example 2: No Interaction (Additive Model) VII

(Historic approach): Fitting the Additive Model

We use the + operator to fit a model assuming the effects of N and P are strictly independent.

Example 2: No Interaction (Additive Model) VIII

```
# Fit additive model (no interaction)
mod_NP_add <- aov(yield ~ N + P, data = NP)
summary(mod_NP_add)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
N            1  189.3  189.28   5.857 0.0247 *
P            1    8.4    8.40   0.260 0.6155
Residuals   21  678.7   32.32
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 2: No Interaction (Additive Model) IX

- **Conclusion:** Nitrogen (`N`) has a significant main effect on yield ($p = 0.0247$), while Phosphate (`P`) does not ($p = 0.6155$). Because the model is additive, we conclude Nitrogen increases yield regardless of whether Phosphate is applied or not.
- We could continue to extract treatment group means and treatment effects using the `emmeans` package as needed (see below).

Extracting Treatment Means and Effects

- Following modern practice, we retain the full model (`mod_NP`) even though the interaction was not significant.
- We can use `emmeans` to extract the estimated marginal means (treatment means) for Nitrogen, averaging over the levels of Phosphate.

Extracting Treatment Means and Effects II

```
library(emmeans)

# Extract Estimated Marginal Means for Nitrogen (N)
# mod_NP is FULL model, not additive
emm_N <- emmeans(mod_NP, ~ N)
emm_N

 N emmean   SE df lower.CL upper.CL
 0   52.1 1.66 20     48.6     55.5
 1   57.7 1.66 20     54.2     61.1

Results are averaged over the levels of: P
Confidence level used: 0.95
```

Extracting Treatment Means and Effects III

Calculating the Treatment Effect

To find the actual size of the treatment effect (the difference between adding Nitrogen vs. not adding Nitrogen), we pass our means into the `pairs()` function.

```
# Test the pairwise difference (Treatment Effect)
pairs(emm_N)
```

```
 contrast estimate   SE df t.ratio p.value
 N0 - N1     -5.62 2.34 20  -2.400  0.0263

Results are averaged over the levels of: P
```

Extracting Treatment Means and Effects IV

- **Interpretation:** The estimated mean yield without Nitrogen is 52.1, and with Nitrogen is 57.7. The `pairs()` output shows that applying Nitrogen increases the mean yield by an estimated 5.62 units, and this main effect is statistically significant.
- Notice the P-value is unchanged from our original ANOVA test.

Factorial Contrasts

Contrasts in Factorial Designs

- Contrasts in a factorial design operate exactly the same way as in a one-way ANOVA. We are simply picking weights to compare means.
- Depending on our research question, we apply these contrasts to two different types of means:
  1. Main Effect Contrasts: We compare the **marginal means** of one factor (averaging over the levels of the other factor).
  2. Cell / Simple Effect Contrasts: We compare the specific **treatment combination means** (μ_{ij}).
- While possible, we do not need to build complex weight matrices by hand. We use software to apply standard adjustments (like Tukey HSD) directly to the means we care about.

Contrasts in Factorial Designs II

Main Effect Contrasts (No Interaction)

If there is no significant interaction, we want to know which levels of Factor A differ from each other overall. We extract the marginal means and apply a Tukey adjustment.

Contrasts in Factorial Designs III

```r
library(emmeans)

# Fit a Three-Way ANOVA
mod <- aov(response ~ Factor_A*Factor_B*Factor_C, data = DATA)

# Extract marginal means for Factor A
emm_A <- emmeans(mod, ~ Factor_A)

# Perform all pairwise comparisons with Tukey adjustment
pairs(emm_A, adjust = "tukey")
```

Contrasts in Factorial Designs IV

Cell Means & Simple Effect Contrasts (Interaction Present)

If an interaction is present, marginal means are misleading. We must compare the specific combinations (cells), or compare the levels of Factor A strictly within a single level of Factor B.

Contrasts in Factorial Designs V

```r
# Extract the means for every specific treatment combination
emm_cells <- emmeans(mod, ~ Factor_A * Factor_B)

# Compare Factor A ONLY within specific levels of Factor B
# (This tests the simple main effects)
contrast(emm_cells, method = "pairwise", by = "Factor_B")

# OR:
# emm_cells <- emmeans(mod, ~ Factor_A | Factor_B)
# contrast(emm_cells, method = "pairwise")

# Alternative: Compare all cells against a specific Control cell

# using Dunnett's adjustment
contrast(emm_cells, method = "trt.vs.ctrl", ref = 1)
```

Advanced Topics in Factorials

Types of Sums of Squares (3-Way Factorial)

- When cell sizes are unequal ($n_{ijk} \neq n$), factors lose orthogonality.
- Why this matters: The sum-of-squares (SS) for a particular term depends on what other terms are included in the model! See Oehlert (Example 10.1 from 2000) for an example.
- This gives rise to three different "types" of SS to consider. In a balanced design, all three give the same answer, so we don't have to worry about the differences.
- The choice of Sums of Squares (SS) dictates which model terms are held constant in the reduced vs. full model comparison ($SS_{\text{Effect}} = SS_{\text{Error, reduced}} - SS_{\text{Error, full}}$).

Types of Sums of Squares (3-Way Factorial) II

- Historically, this was very problematic, and caused people to avoid analyzing data from unbalanced factorial designs. Today, software can readily solve this issue for us.

Effect	Type I (Sequential)	Type II (Hierarchical)	Type III
A	$SS(A)$	$SS(A \mid B, C)$	$SS(A \mid$
B	$SS(B \mid A)$	$SS(B \mid A, C)$	$SS(B \mid$
C	$SS(C \mid A, B)$	$SS(C \mid A, B)$	$SS(C \mid$
AB	$SS(AB \mid A, B, C)$	$SS(AB \mid A, B, C, AC, BC)$	$SS(AB$
AC	$SS(AC \mid A, B, C, AB)$	$SS(AC \mid A, B, C, AB, BC)$	$SS(AC$
BC	$SS(BC \mid A, B, C, AB, AC)$	$SS(BC \mid A, B, C, AB, AC)$	$SS(BC$
ABC	$SS(ABC \mid A, B, C, AB, AC, BC)$	$SS(ABC \mid A, B, C, AB, AC, BC)$	$SS(AB$

Table 1: Comparison of Sums of Squares conditional notation for factors A, B, and C.

Type I Sum of Squares

Type I (Sequential)

Summary: The SS for any term is adjusted *only* for the terms that precede it in the model.

- For historic reasons, this is the output of the ‘aov()‘ function.
- The hypothesis being tested is whether adding a specific term improves the model, given the terms already present.
- **Disadvantage:** The SS depends on the order in which you specify model terms!

Type I Sum of Squares II

Which Type Should You Use in Practice?

- **Type I:** Avoid unless testing a strict hierarchical sequence (e.g., covariate first). Order of terms in `aov()` changes the output!
- **Type II:** Best default when you are confident higher-order interactions are zero ($AB = 0$). Maximizes statistical power.
- **Type III:** Standard for factorial experiments. Tests equal weighting across all treatment cells regardless of sample size differences.

Unbalanced Data: Practical Workflow in R

- Base R's `summary(aov())` defaults to Type I SS, making it dangerous for unbalanced data.
- **Solution:** Use `joint_tests()` from `emmeans`. It provides Type III hypothesis tests safely without requiring you to change R's global options.

```r
library(emmeans)

# Fit standard model on unbalanced data
mod_unbal <- aov(yield ~ P * K, data = unbalanced_data)

# 1. Run Type III ANOVA tests safely
joint_tests(mod_unbal)
```

Unbalanced Data: Practical Workflow in R II

Practical Decision Rules

1. **Look at the Interaction p-value first:**
    - If $p \geq 0.05$ (No interaction): Interpret overall main effects and run overall pairwise comparisons: `emmeans(mod, P)`.
    - If $p < 0.05$ (Interaction present): Stop! Do not interpret global main effects. You must analyze **Simple Main Effects**.

Analyzing Simple Main Effects

- Definition: A **simple main effect** tests the effect of Factor A at *one specific fixed level* of Factor B.
- When an interaction exists, stating "Superphosphate increases yield" is incomplete. You must state "Superphosphate increases yield *when Potash is absent*, but has no effect *when Potash is present*."
- Use the `by` argument in `emmeans` to slice the analysis.

Analyzing Simple Main Effects II

```r
# Obtain EMMs for the interaction
emm_int <- emmeans(mod_unbal, ~ P * K)

# Test the simple main effect of Potash (K) WITHIN each level of
contrast(emm_int, method = "pairwise", by = "P")

# Plot simple effects visually
emmip(mod_unbal, K ~ P)
```

Multiple Comparisons: Beyond the F-Test

Multiple Comparisons in Practice

- A significant ANOVA interaction tells you *some* difference exists, but testing all pairs blindly increases your family-wise error rate (false positive rate).
- The `emmeans` package provides a streamlined pipeline for post-hoc testing with appropriate data-dredging penalties.

```
# Assuming fert_mod is our fitted two-way model
emm_int <- emmeans(fert_mod, ~ P * K)

# Tukey adjustment for all pairwise comparisons
pairs(emm_int, adjust = "tukey")
```

Multiple Comparisons in Practice II

Visualizing Contrasts

- Staring at massive tables of p-values makes it difficult to assess effect sizes.
- You can pass your contrast object directly into a plot to visually see which adjusted confidence intervals exclude zero.

```
# Generates a forest plot of the differences between groups
plot(pairs(emm_int, adjust = "tukey")) +
  ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color
  ggplot2::labs(title = "Tukey-Adjusted Pairwise Differences")
```

Resource Constraints: Unreplicated & Fractional Designs

Unreplicated Factorials ($n = 1$)

- Experiments are expensive. Sometimes we can only afford a single replicate per treatment combination ($n = 1$).
- The Problem: Fitting a full factorial model uses up all available degrees of freedom. The error degrees of freedom (df_E) becomes 0, so R returns `NA` for all standard errors and p-values.

```
# Fits main effects, 2-way, 3-way, and 4-way interactions
mod_unrep <- aov(yield ~ A * B * C * D, data = unrep_data)

# F-values and p-values will return as NA
summary(mod_unrep)
```

Unreplicated Factorials ($n = 1$) II

The Fix: Pooling into Error

- Relying on the Sparsity of Effects Principle, we assume high-order interactions (e.g., 3-way and 4-way) are effectively zero (just random noise).
- We restrict the model, pushing those negligible higher-order terms into the residual error to mathematically generate degrees of freedom.

Unreplicated Factorials ($n = 1$) III

```r
# Restrict the model to main effects and 2-way interactions only
# The "^2" syntax drops 3-way and 4-way terms into the error
mod_pooled <- aov(yield ~ (A + B + C + D)^2, data = unrep_data)

# Now R can calculate p-values for main and 2-way effects!
summary(mod_pooled)
```

Fractional Factorials & Confounding

- A fractional factorial design runs only a structured fraction of the full combinations (e.g., 16 runs instead of 32 for a 5-factor design).
- **How R handles it:** R automatically recognizes the missing runs and returns `NA` for the interaction terms it cannot mathematically estimate.
- **Aliasing:** Because we skipped runs, certain effects are perfectly correlated (confounded).
    - *Intuition:* If we design a half-fraction by defining Factor C's levels mathematically as $A \times B$, R cannot tell if an observed change in yield is caused by the main effect of C, or the interaction of A and B.

Fractional Factorials & Confounding II

```r
# Generate a mock half-fraction dataset (2^(3-1))
library(FrF2)
frac_data <- FrF2(nruns = 4, nfactors = 3, factor.names = c("A",
frac_data$yield <- c(14, 18, 22, 19)

frac_mod <- aov(yield ~ A * B * C, data = frac_data)

# This outputs the exact matrix showing which main effects
# are mathematically identical to which interaction terms:
alias(frac_mod)
```

```
Model :
yield ~ A * B * C

Complete :
          (Intercept) A1 B1 C1
A1:B1     0           0  0  1
```

References and Acknowledgements

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.

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